

Inverse Scattering Solver Based on Deep Neural Network With Total Variation Regularization

Jie Ma, Zicheng Liu , and Yali Zong 

Abstract—Deep neural network (DNN) techniques have been applied to solve nonlinear electromagnetic inverse scattering problems (ISP) and shown potentials in superior imaging performances over traditional methods. In DNN ISP solvers, data features are learned from imaging samples to accurately represent the contrast distribution through the optimization of hyperparameters of neural networks, while the loss function, which often quantifies discrepancies between the predicted imaging result and the ground truth, is targeted to be minimized in the training process. In this letter, to further make use of the (piecewise) smoothness of the contrast distribution, the desired solution is regularized by additionally penalizing its total variation (TV) in the loss function. Effects of the TV regularization on the DNN ISP solver are studied and summarized with various simulation settings, and the improvement in imaging performance is verified based on the experimental data.

Index Terms—Deep neural network (DNN), inverse scattering problem (ISP), loss function, regularization, total variation (TV).

I. INTRODUCTION

ELECTROMAGNETIC inverse scattering is the imaging technique reconstructing the distribution of relative permittivity or conductivity of the region in detection based on the collected scattered fields [1]. The advantages of noninvasive imaging lead to the wide application in ground penetrating radar [2], nondestructive testing (NDT) [3], and biomedical imaging [4]. Super-resolution imaging can be achieved by considering multiple scattering behaviors in scatters, which make inverse scattering problems (ISPs) nonlinear and possibly stuck in local optima.

Traditional methods for solving ISPs can be divided into two categories. Deterministic methods include distorted Born iterative method [5], contrast source inversion [6], and subspace-based optimization method [7]. Stochastic methods include genetic optimization [8], differential evolution [9], and particle swarm optimization [10]. To overcome the ill-posedness of ISPs [11], regularization methods are widely used to explore the prior information of the desired solution and, due to the nonlinearity, the optimal solution is often obtained with iterative schemes in which the forward modeling problem is solved at

each iteration leading to high computational cost that prevents the real-time application of traditional ISP solvers.

A deep neural network (DNN) has been successfully applied in image processing [12], computer vision [13], nature language processing [14], and electromagnetic computation [15], [16]. In the learning-by-examples paradigm, scattered electric fields are taken as input of the surrogate model, which can estimate defect parameters in NDT [17], [18]. To incorporate imaging physics in the training process, the domain of interest (DOI) is first imaged with a fast traditional method [e.g., back propagation (BP)] and the obtained image is then input to a sequence of convolutional neural network (CNN) modules to have refined result [19], [20], [21], [22], [23]. To decrease the nonlinearity of the involved objective function, an inverse source problem is solved based on DNN to retrieve induced current and the contrast distribution is then obtained with a deterministic scheme [24].

Incorporating prior knowledge about the desired solution, such as sparsity and low-rank property, has been shown having the potential to improve the imaging performances [25], [26], [27]. Such information is usually used through the penalization of the associated losses, which should be minimized and retrieved following an iterative scheme. In this letter, the effects of regularization are studied based on a U-net DNN solver. Concerning scatterers with (elementwise) smoothly varied permittivity distribution, the total variation (TV) of the predicted contrast distribution is expected to be sparse, i.e., a small portion of elements being nonzero or large valued. Effects of TV regularization in the DNN solver are discussed based on simulated datasets, and the improved imaging performance is verified by an experimental testing.

The rest of this letter is organized as follows. Section II introduces the electromagnetic inverse scattering theory. Section III presents the loss function with TV regularization and the network settings. Numerical results are given in Section IV. Finally, Section V concludes this letter.

II. ELECTROMAGNETIC ISP

As shown in Fig. 1, 2-D nonlinear electromagnetic ISPs are concerned where the illumination wave is transverse-magnetic (TM) polarized and the scatterers within DOI are invariant along the z -axis. Such conditions lead to scalar scattering problems, which are described by the state equation

$$E_{\text{DOI}}^{\text{tot}}(\mathbf{r}) = E_{\text{DOI}}^{\text{inc}}(\mathbf{r}) + k_0^2 \iint_D g(\mathbf{r}, \mathbf{r}') J(\mathbf{r}') d\mathbf{r}', \mathbf{r} \in D \quad (1)$$

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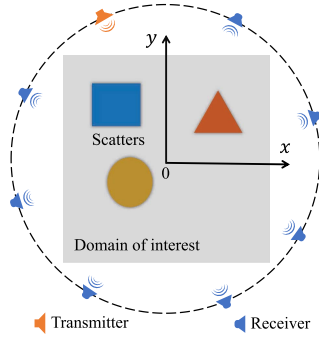


Fig. 1. Sketch of the concerned 2-D electromagnetic ISP based on the information collected with multiple emitting source and receivers.

and the data equation

$$E^{\text{sca}}(\mathbf{r}) = k_0^2 \iint_D g(\mathbf{r}, \mathbf{r}') J(\mathbf{r}') d\mathbf{r}', \quad \mathbf{r} \in S \quad (2)$$

which quantify the multiple scattering behaviors within DOI and scattered fields when the observation position $\mathbf{r}$ is in the antenna region (outside DOI), respectively. The superscripts “tot,” “inc,” and “sca,” respectively, denote the total field, incident field, and scattered field. $g(\mathbf{r}, \mathbf{r}') = iH_0^{(1)}(k_0|\mathbf{r} - \mathbf{r}'|)/4$ is the scalar Green’s function, where k_0 is the wave number of the background medium, and $H_0^{(1)}$ is the first-kind zero-order Hankel function. J stands for the contrast source with the definition

$$J(\mathbf{r}') = \chi(\mathbf{r}') E_{\text{DOI}}^{\text{tot}}(\mathbf{r}'). \quad (3)$$

The contrast $\chi(\mathbf{r}') = \varepsilon_r(\mathbf{r}') - 1$, $\varepsilon_r(\mathbf{r}')$ being the relative permittivity at $\mathbf{r}'$. The air background is assumed here and scatterers are nonmagnetic. The total field $E_{\text{DOI}}^{\text{tot}}$ is a function of contrast, which makes the retrieval based on E^{sca} a nonlinear problem.

The image of scatterers is obtained based on the field data collected by alternatively illuminating the DOI by transmitting antennas and at each time recording the scattered fields at positions of receivers. The DNN solver presented in this letter is trained based on the dataset from the simulated scattered fields where the method of moment (MoM) is applied with the pulse basis function and the delta testing function.

III. DNN SOLVER

A. U-Net CNN Architecture

The structure of the U-net CNN [28] is sketched in Fig. 2, which consists of an encoding (down-sampling) path and a decoding (up-sampling) path. Both paths include repeated convolution layers with 3×3 kernels, batch normalization (BN), and rectified linear unit (ReLU). In the encoding path, when the dimension of feature map is halved using maximum pooling, the channel number is doubled. The decoding part follows the contrary procedures except that after doubling the spatial dimension using 3×3 up-convolutions, BN and ReLU activation are performed. The jump connection between the two paths can preserve the information of features in different depths and improve the efficiency in representing the imaged scatterers based on the learned features.

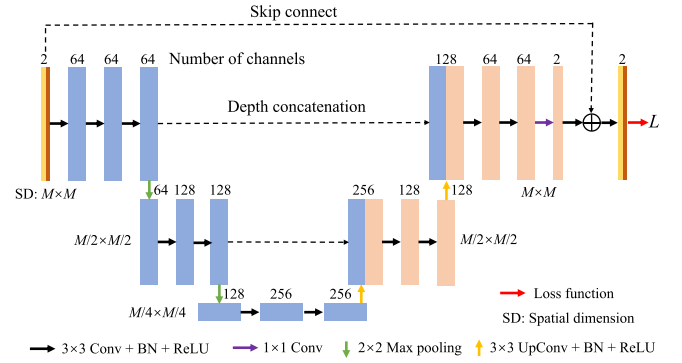


Fig. 2. U-net architecture used to train ISP solvers. The input has two channels for real part and imaginary part of the initial estimate of contrast, and BN denotes batch normalization.

The estimate of the contrast function based on the fast traditional ISP solver BP [28] is taken as input. Treating the real part and the imaginary part as two independent channels, the complex-valued inversion problem is solved using the real-valued neural network.

B. Loss Function

The Euclidean distance between the predicted and the ground truth is often penalized in the training process of DNN solvers. Since the scatterers usually have smooth or piecewise smooth permittivity distribution, to further utilize this prior information, the loss function, which additionally constrains the TV of the response, is applied, i.e.,

$$L(\hat{\chi}) = \|\hat{\chi} - \chi\|_2^2 + \frac{\beta}{2} \sum_{i,j} \left[(\chi_{i,j-1} - \chi_{i,j})^2 + (\chi_{i+1,j} - \chi_{i,j})^2 \right] \quad (4)$$

where the first part quantifies the discrepancies of the response to the ground truth and the second imposes the constraint from TV regularization. $\chi_{i,j}$ denotes the contrast of the grid located at the i th row and the j th column. For piecewise scatterers, the second part would yield a sparse matrix, the nonzero elements of which indicate the boundary of scatterers. The regularization parameter β balances the weights from the two parts. When β is 0, there is no regularization term in the loss function. As β increases gradually, the weight of the regularization term in the loss function also increases.

C. Training Dataset

The training dataset is obtained based on the simulations where the DOI is sized by $5.6\lambda_0 \times 5.6\lambda_0$ and divided into 64×64 grid cells. λ_0 is the working wavelength in vacuum and equals 7.5cm. In total, 36 linearly polarized transmitters and the same number of receivers are evenly distributed on a circle, which shares the center of DOI and the radius $10\lambda_0$.

Synthetic homogeneous digit-like scatterers are generated based on the handwritten digit database MNIST [29] by randomly setting the value of relative permittivity in the range of [1, 5] to consider low- and high-nonlinear ISPs. Computing the

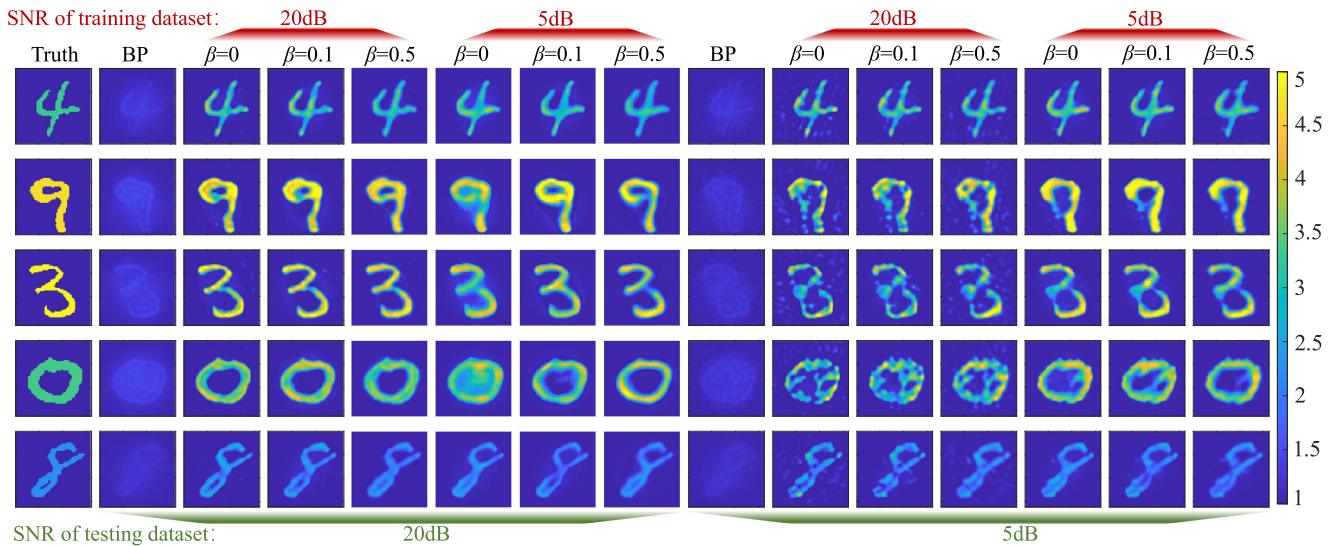


Fig. 3. Testing over digit-like scatterers with the DNN solvers trained based on the scattered fields with different noise levels and different values of β .

scattered field collected by receivers with MoM and simulating the field contamination as additive Gaussian noise, the input is obtained with BP. Although the scatterers are assumed lossless, complex-valued contrast distributions are obtained with BP. To solve complex-valued ISP by a real-valued neural network, the input layer has two independent channels, which allow the real part and the imaging part of BP contrast distribution to be fully considered and also make the proposed DNN solver could be straightforwardly generalized to lossy scatterers.

D. Training Settings

The DNN solver is trained with MATLAB 2022b on a workstation with 1.67TB memory, 2.1 GHz CPU, and NVIDIA RTX 6000 graphics card. Stochastic gradient descent with momentum 0.99 is applied for the optimization of the network hyperparameters (weights and biases). The learning rate is initialized as 5×10^{-6} and halved every 20 epochs, whereas the learning process was stopped when the epoch number reached 150. The training dataset includes 1000 digit-like images and about 15 min are required to complete the training process.

IV. RESULTS

The performance of the obtained DNN solvers is demonstrated by predicting digit-like scatterers (in the testing dataset), polygon-like scatterers, scatterers with the Austria profile, and finally verified based on the experimental data from Institute Fresnel [30].

A. Testing Over Digit-Like Scatterers

Fig. 3 shows representative examples of imaging digit-like scatterers where the distribution of retrieved amplitude of complex relative permittivity is presented. The tests are performed with different values of β and effects of signal-to-noise ratio (SNR) mismatching are studied. Although the input of a neural network, i.e., BP imaging result, tends to underestimate the

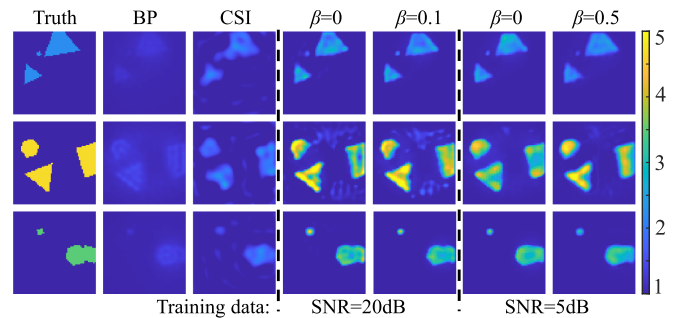


Fig. 4. Testing over polygon-like scatterers when the testing dataset is generated with scattered fields with SNR = 20 dB.

relative permittivity, significant improvements are observed by the DNN solvers in terms of imaging accuracy. Moreover, when the SNR of the training dataset equals 20 dB, $\beta = 0.1$ seems a better option considering less artifacts in the background, despite the fact when predicating a 5 dB testing dataset, strong artifacts appear and the digit number even cannot be clearly recognized. However, changing the noise level of the training dataset as 5 dB seems can alleviate the SNR mismatching effects. Testing on the dataset with SNR = 20 dB, although the imaging performance of the DNN solver trained with SNR = 5 dB is still inferior to the one with SNR = 20 dB, the geometric feature and the value of relative permittivity can be well reconstructed except that the optimal value of β is increased to 0.5, which indicates that when SNR is low, incorporating the prior information about scatterers can effectively improve the imaging performance and more weights should be given to the regularization term in the loss function.

B. Testing Over Polygon-Like Scatterers

The polygon-like scatterers are generated by randomly positioning polygons with 3–7 sides inside DOI and the relative permittivity of the scatterers is in the range of [1, 5]. Fig. 4 shows

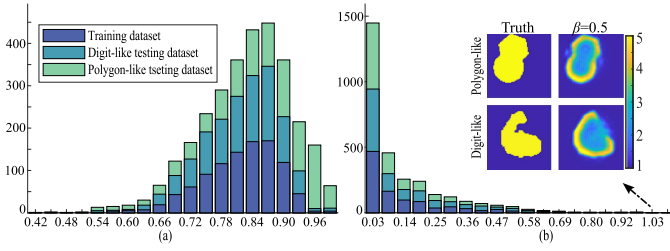


Fig. 5. Histograms of (a) SSIM and (b) MSE when imaging samples in the training dataset, digit-like, and polygon-like scatterers in the testing dataset.

TABLE I
STATISTICAL PARAMETERS FOR TEST RESULTS OF DIGIT-LIKE AND
POLYGON-LIKE SCATTERERS

	Training dataset	Digit-like	Polygon-like
MSE mean	0.1269	0.1274	0.1218
MSE variance	0.0234	0.0270	0.0262
SSIM mean	0.8166	0.8154	0.8515
SSIM variance	0.0052	0.0058	0.0107

the examples of imaging polygon scatterers when the scattered fields are contaminated by 20 dB Gaussian noise. When the network is trained based on the dataset at the same noise level, the improved reconstruction accuracy is achieved by imposing the smoothness on scatterers with $\beta = 0.1$. When the SNR of training dataset equals 5 dB, noise effects are effectively learned by the networks and the artifacts in the background are cleared. Setting β as 0.5 leads to reduced bias on reconstructing relative permittivity. The testing results again indicate that the obtained DNN ISP solver, which is trained based on the dataset with low SNR, is robust to SNR mismatching when applying the solver in higher SNR scenarios.

Fig. 5 and Table I present the statistical analysis based on prediction of samples in the training dataset, 1000 digit-like, and 1000 polygon-like testing scatterers, respectively. The network is trained based on digit-like scatterers with SNR = 5 dB, and β is set as 0.5. Mean squared error (MSE) and structure similarity index measure (SSIM) are used for quantifying the imaging performances. As seen, samples for training and the same type of samples for testing can be predicted with almost the same accuracy. Fig. 5(b) shows the imaging examples with large values of MSE which reveals that when the relative permittivity of the scatterer is 5, biased estimation can be obtained due to sparse sampling at the boundary of the relative-permittivity range.

C. Testing Over Scatterers With Austria Profile

The performance of the obtained DNN solvers for ISPs with different levels of nonlinearity is tested with scatterers with the Austria profile. As shown in Fig. 6, the profile is made of two circular disks and an annulus, the relative permittivity of which varies from 2 to 5. With larger contrast, the mutual scatterings between scatterers challenge the accuracy in retrieving the geometric feature. When $\epsilon_r = 5$, the images of the disks and the annulus even have intersections.

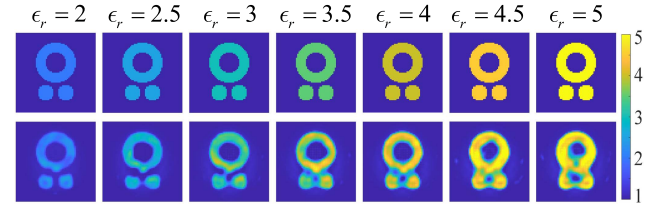


Fig. 6. Testing over scatterers with the Austria profile, training dataset with SNR = 5 dB and $\beta = 0.5$.

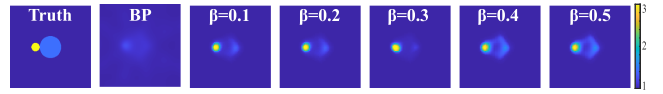


Fig. 7. Imaging two cylinders based on the “FoamDielExt” experimental data, and the training dataset with SNR = 5 dB.

D. Testing Over Experimental Data

The imaging performance is verified based on the experimental data in [30] named as “FoamDielExt,” which consists of two homogeneous cylinders, where the one with diameter 80 mm has the relative permittivity of 1.45 ± 0.15 and the small one with diameter 31 mm has the relative permittivity of 3 ± 0.3 . In total, eight transmitters and 241 receivers are used to collect scattered fields with TM polarization at the frequency of 4 GHz. The working wavelength and the DOI size are the same with the settings for the training dataset. Inputting the highly biased estimated contrast distribution from BP to the DNN solvers, significant improvements are observed from Fig. 7 when imaging with the DNN solvers. The geometric feature of the weak scatterer is enhanced when the TV loss is penalized, although errors still happen especially in reconstructing the circular shape of the weak scatterer.

V. CONCLUSION

The ISP solvers based on U-net DNN with the TV regularization is obtained and studied in this letter. Although the input from BP is biased, the DNN ISP solvers yield imaging results with significant improvements. Benefiting from penalizing the TV regularization term in the loss function, the obtained solvers enjoy the enhancement of geometric features and the suppression of background artifacts. The imaging performance and the generalization ability of the obtained DNN solvers are tested over digit-like, polygon-like scatterers, scatterers with the Austria profile, and the two cylinders in experiment. When DNN solvers are trained based on low-SNR data, the obtained ISP solver is more robust to noise effects. While the analysis shows the improvement in imaging performances with the TV regularization, it also discloses that the TV regularization may help little in accurately reconstructing weak scatterers in the presence of strong neighbor scatterers.

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